

4A. MANUAL CALCULATION OF BRANCH-NODE INCIDENCE MATRIX

GIVEN DATA

Number of Nodes = n

Number of Branches = b

For each branch and node:

- If the branch is connected and directed towards the node, assign +1.
- If the branch is connected and directed away from the node, assign -1.
- If the branch is not connected to the node, assign 0.

SOLUTION

STEP 1 : IDENTIFICATION OF BRANCH CONNECTIONS

For every branch, determine whether it is connected to a node.

Case 1:

Connected and directed towards the node

$$K(i,j) = +1$$

Case 2:

Connected and directed away from the node

$$K(i,j) = -1$$

Case 3:

Not connected

$$K(i,j) = 0$$

STEP 2 : FORMATION OF BRANCH-NODE INCIDENCE MATRIX

Suppose there are 4 nodes and 3 branches.

Branch 1 : Node 1 $\rightarrow$ Node 2

Branch 2 : Node 2 $\rightarrow$ Node 3

Branch 3 : Node 3 $\rightarrow$ Node 4

For Branch 1

Node 1 = -1

Node 2 = +1

Node 3 = 0

Node 4 = 0

For Branch 2

Node 1 = 0

Node 2 = -1

Node 3 = +1

Node 4 = 0

For Branch 3

Node 1 = 0

Node 2 = 0

Node 3 = -1

Node 4 = +1

FINAL BRANCH-NODE INCIDENCE MATRIX

[-1 1 0 0]

[0 -1 1 0]

[0 0 -1 1]

4B. MANUAL CALCULATION OF CUTSET MATRIX

GIVEN DATA

Number of Elements (Branches) = 6

Number of Cutsets = 5

The presence and orientation of each element with respect to the cutsets are considered as follows:

- If the element is present and directed along the cutset, assign +1.
- If the element is present and directed opposite to the cutset, assign -1.
- If the element is absent from the cutset, assign 0.

SOLUTION

STEP 1 : IDENTIFICATION OF ELEMENTS IN CUTSETS

Element 1 is present only in Cutset 1 and directed along the cutset.

Therefore,

$A_{11} = 1$

$$A_{12} = A_{13} = A_{14} = A_{15} = 0$$

Hence,

$$\text{Element 1} \rightarrow [1 \ 0 \ 0 \ 0 \ 0]$$

Element 2 is present only in Cutset 2 and directed along the cutset.

Hence,

$$\text{Element 2} \rightarrow [0 \ 1 \ 0 \ 0 \ 0]$$

Element 3 is present only in Cutset 3 and directed along the cutset.

Hence,

$$\text{Element 3} \rightarrow [0 \ 0 \ 1 \ 0 \ 0]$$

Element 4 is present only in Cutset 4 and directed along the cutset.

Hence,

$$\text{Element 4} \rightarrow [0 \ 0 \ 0 \ 1 \ 0]$$

Element 5 is present only in Cutset 5 and directed along the cutset.

Hence,

$$\text{Element 5} \rightarrow [0 \ 0 \ 0 \ 0 \ 1]$$

Element 6 is present in Cutset 2 and directed along the cutset and is present in Cutset 4 but directed opposite to the cutset.

Therefore,

$$A_{62} = 1$$

$$A_{64} = -1$$

Remaining elements are zero.

Hence,

$$\text{Element 6} \rightarrow [0 \ 1 \ 0 \ -1 \ 0]$$

FINAL CUTSET MATRIX

$$[1 \ 0 \ 0 \ 0 \ 0]$$

$$[0 \ 1 \ 0 \ 0 \ 0]$$

$$[0 \ 0 \ 1 \ 0 \ 0]$$

$$[0 \ 0 \ 0 \ 1 \ 0]$$

$$[0 \ 0 \ 0 \ 0 \ 1]$$

$$[0 \ 1 \ 0 \ -1 \ 0]$$